

Predeclared Analysis Protocol

Changing connections: temporal potential structural connectivity, unprotected pinch points, and monitoring priorities for free-ranging cheetahs in southern Africa

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What this document is. A complete statement of the analytical choices for this project, fixed before any connectivity output is generated or inspected. Its purpose is to remove researcher degrees of freedom from a study design that contains a very large number of individually defensible options. Every value below was chosen without reference to any result.

What it is not. It is not immutable. Changes are permitted and expected; they are recorded in the amendment log (§20) with a date and a reason, and the log is published with the manuscript. A disclosed amendment is sound practice. A silent change is not.

1 Document control and sign-off

Protocol version	0.1 (draft)
Date drafted	October 5, 2026
Date locked	_____
Author	Liam Crettol
Author signature	_____
Git lock tag	<code>protocol-v1.0</code> commit: _____
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Amendments to date	0

The lock recorded above attests that this protocol was fixed before any connectivity, corridor, current-flow, protected-area overlap, or prioritisation result had been generated or viewed. Because this is single-author work with no countersignature, the verifiable evidence of the lock date is external: the signed git tag and the OSF timestamp, both of which are dated independently of the author.

2 Purpose and contribution

This project models how potential structural connectivity between established cheetah range cores has changed across southern Africa between 2012 and 2024, identifies which pinch points remain important across plausible modelling assumptions, characterises their protected-area status, and derives a ranked set of monitoring priorities.

Two boundaries define the contribution and both are binding.

1. **This is not a habitat suitability study.** Regional current and projected cheetah habitat suitability, including protected-area mismatch, is already published. Suitability may enter this analysis only as an input to a resistance scenario, never as a result.
2. **Range cores are held fixed at the historical baseline.** No comparable range reassessment exists after the 2010–2016 evidence period. Cores therefore do not move between snapshots, and

all measured temporal change is landscape-side. This must be stated in the abstract, not only the methods.

3 Objectives and hypotheses

3.1 Objectives

- O1. Quantify change in modelled potential structural connectivity between fixed range cores across four snapshots spanning 2012–2024.
- O2. Identify pinch points that remain important across the full sensitivity grid, and characterise their protected-area status.
- O3. Derive and stress-test a ranked set of monitoring priorities.

3.2 Hypotheses as locked

H1 (revised from the workbook). Connectivity loss is concentrated on a minority of network links rather than distributed across the network, and the affected links are disproportionately those with high betweenness in the baseline network.

Note on the revision. H1 as originally drafted predicted that cells with larger increases in human pressure would show larger increases in resistance. That is true by construction, since the pressure layers are the resistance inputs. It cannot fail and therefore is not a hypothesis. The revised form asks a question the model can answer negatively: pressure change might fall on redundant links, in which case network-level connectivity is robust to it and the conservation implication reverses.

Refuted if: connectivity loss is approximately uniform across links, or loss falls preferentially on low-betweenness links.

H2. High-value connectivity area is disproportionately located outside formal protected areas.

Refuted if: the outside-PA share of high-current area does not exceed the predeclared null (§13) at the stated significance level, or the result does not hold across current-flow thresholds.

H3. A limited subset of transboundary linkages remains important across all resistance assumptions, while other apparent corridors are strongly model-sensitive.

Refuted if: agreement is uniformly high (in which case the ensemble adds nothing) or uniformly low (in which case no robust set exists and the paper’s primary output becomes the uncertainty map, per the decision rule in §18).

H4. The highest-priority monitoring cells occur where structural importance coincides with old or sparse cheetah evidence and rapid landscape change.

Refuted if: no stable top tier persists across the predeclared weight sets.

H1 revision confirmed. The original form was true by construction (pressure layers are the resistance inputs) and could not fail, so it was not a hypothesis. The revised form above is retained as stated.

4 Study extent and exclusions

Primary extent. The confirmed and likely free-ranging cheetah range envelope across Namibia, Botswana, Zimbabwe and South Africa, as delineated in the archived 2010–2016 regional range assessment, buffered outward by 100 km to avoid truncating corridors at the envelope edge.

Rationale for the buffer. Least-cost and current-flow results are sensitive to boundary truncation. A buffer wider than the plausible inter-core path deviation prevents the analysis boundary from

becoming an artificial barrier. Cores are not defined in the buffer; it exists only so paths can route through it.

Exclusions. Permanent water bodies; cells with no valid data in any covariate for any snapshot; areas outside the four primary countries except where inside the buffer.

Scope-control fallback. If total compute for the full sensitivity grid exceeds **120 hours** on available hardware, the extent reduces to the Namibia–Botswana–Zimbabwe transboundary network. This trigger is numeric and declared in advance so that a scope reduction is a planned contingency rather than a response to inconvenient results. South Africa would then be reported as a validation region only.

100 km buffer confirmed. It is the same order of magnitude as the individual movement-distance value adopted for the PC dispersal parameter below, which gives the two figures internal consistency, and it is wide enough that corridor routing is not truncated at the boundary.

DECISION REQUIRED: The 120-hour compute trigger remains provisional pending a pilot run on the development hardware (GitHub Codespace: 4-core, 16GB RAM, 32GB storage). Time one full sensitivity-grid unit run (one resistance scenario $\times$ one core definition $\times$ one snapshot at 1 km) end to end in Circuitscape.jl pairwise mode, multiply by 144 runs, and compare the result against the 120-hour ceiling before timeline block 5. Amend this value with the measured figure once the pilot is run.

5 Coordinate system and working resolution

Coordinate reference system. Africa Albers Equal Area Conic (ESRI:102022), an equal-area projection appropriate for continental Africa. Equal-area is required because the analysis reports areal quantities (protected area shares, corridor areas, tabulated change).

Full WKT confirmed below (Esri’s standard definition for ESRI:102022).

```
PROJCS["Africa_Albers_Equal_Area_Conic",GEOGCS["WGS 84",DATUM["WGS_1984",SPHEROID["WGS 84",6378137,298.257223563,AUTHORITY["EPSG","7030"]],AUTHORITY["EPSG","6326"]],PRIMEM["Greenwich",0,AUTHORITY["EPSG","8901"]]
```

Working resolution. 1 km, with the full pipeline repeated at 2 km and 5 km as resolution sensitivity.

Rationale. The occurrence baseline is a 10 km assessment grid; environmental covariates range from 30 m to 1 km. Modelling at 10 km would make a “pinch point” a 100 km² block, which is not an actionable location and not a pinch point in any meaningful sense. Modelling at 100 m would imply locational precision the occurrence evidence cannot support and would raise compute by two orders of magnitude. 1 km is the coarsest resolution at which a pinch point is a place a survey team could go, and the finest at which the claim remains honest. The two coarser repetitions demonstrate that conclusions are not resolution artefacts.

Alignment. All rasters share a single snap raster, origin, cell size, extent and NoData convention. Aggregation from finer sources uses mean for continuous variables and majority for categorical variables; both are recorded per layer.

6 Temporal design

Snapshots. 2012, 2016, 2020, 2024.

Nearest-year rule. Where a covariate is not available for a snapshot year, the nearest available year is used and the offset in years is recorded. The temporal alignment table (one row per covariate,

one column per snapshot, each cell giving source year and offset) is a required deliverable and is published in the supplement.

Maximum tolerated offset. ± 2 years. A covariate that cannot meet this for a given snapshot is treated as static for the entire analysis rather than being interpolated. Interpolation is prohibited: it would create apparent change from an assumption rather than an observation.

7 Range cores

Three core definitions, all derived from the archived confirmed-range data, applied identically to every snapshot.

ID	Evidence included	Minimum patch area	Character
C1	Confirmed range only	2000 km ²	Conservative
C2	Confirmed + likely range	1000 km ²	Moderate (default for headline figures)
C3	Confirmed + likely + possible	500 km ²	Inclusive

Rationale for the area thresholds. Cheetah home ranges on southern African farmland are reported at the order of hundreds to over a thousand square kilometres. A core smaller than roughly one home range is not a population source in any useful sense. The three thresholds bracket that magnitude rather than asserting a precise value.

Area thresholds confirmed. Marker et al. 2008 report a mean cheetah home range of 1651 km² (± 1594 km²) on north-central Namibian farmland; Melzheimer et al. 2020 report 380 km² for territory holders, up to 1600 km² for floaters spanning multiple territories, and 650 km² for females. This confirms the “hundreds to over a thousand” order of magnitude above, and the three thresholds bracket it without adjustment.

Amendment: evidence category mapping. The archived Weise et al. 2017 Dryad raster (`cheetahsouthernafrica`) does not use confirmed/likely/possible labels. Its raster attribute table instead codes: Value 1 = Possible Cheetah Presence; Value 2 = Free Range Cheetah Presence Buffer; Value 3 = Free Range Cheetah Presence; Value 4 = Fenced Cheetah Presence. Mapped onto the core definitions above: C1 (confirmed range only) uses Value 3 only; C2 (confirmed + likely) adds Value 2; C3 (confirmed + likely + possible) adds Value 1. Value 4 (fenced populations) is excluded from all three core definitions, consistent with the free-ranging scope of this study.

Cores are generated by region grouping on the thresholded evidence surface, with a single morphological closing pass to remove single-cell holes. No manual editing of core geometry is permitted at any stage.

8 Covariate register and temporal treatment

Every covariate is classified as **dynamic** (varies across snapshots and therefore carries temporal signal) or **static** (identical in every snapshot). This classification is locked here and cannot change after results are seen.

Covariate	Source	Treatment	Note
Built-up fraction	GHSL	Dynamic	Epochal; apply nearest-year rule
Nighttime radiance	VIIRS annual	Dynamic	Mask known gas flares before differencing
Percent tree cover	MODIS VCF	Dynamic	Pin one collection version across all years
Percent non-tree vegetation	MODIS VCF	Dynamic	As above
Percent bare ground	MODIS VCF	Dynamic	As above
Land cover (categorical)	MODIS MCD12Q1	Static	Context and masking only; never differenced
Roads	GRIP4 (primary)	Static	OSM for contemporary detail only
Livestock density	GLW4	Static	Single reference year; exposure only
Veterinary fences	national / AHEAD	Static	See §8.1
Distance to water	HydroSHEDS	Static	
Slope / ruggedness	SRTM	Static	
Protected areas	WDPA	Static	Overlay only, not a resistance input

Why categorical land cover is static. Annual categorical land-cover products reassign classes between adjacent years for reasons unrelated to landscape change, and the instability concentrates in exactly the semi-arid mixed grass and shrub classes that dominate this study area. Differencing them would manufacture change. Temporal vegetation signal is therefore carried entirely on the continuous VCF fields.

8.1 Veterinary fences

Veterinary cordon fences in Botswana and Namibia are major deliberate barriers to large mammal movement and must be represented. Acquisition proceeds in this order: national veterinary department layers; the AHEAD/WCS KAZA fence assessment materials; OSM **barrier=fence** where mapped.

Contingency. If no usable vector layer can be obtained by the end of timeline block 3, fences are handled as an explicit scenario comparison (network modelled with and without digitised fence lines from the best available source) and the limitation is stated prominently. Omitting fences silently is not an option.

Acquisition order and contingency trigger confirmed. National veterinary department layers rank first as the most authoritative source, the AHEAD/WCS KAZA fence assessment second, and OSM **barrier=fence** last as a crowdsourced fallback. The trigger is the end of timeline block 3 on the Master Workflow schedule, not a calendar date: if the acquisition sequence above has not produced a usable vector layer by that point in the workflow, the scenario-comparison contingency above is invoked automatically.

9 Resistance scenarios

Six scenarios. RES-01 to RES-05 are as specified in the project workbook's Resistance Scenario Register, reproduced by reference and treated as locked. RES-06 is added here.

RES-06 (composite pressure). Resistance driven by a single published composite human-modification index rather than by separately weighted pressure layers. Its purpose is to test whether

conclusions depend on the analyst’s own weighting of correlated human-pressure variables, which is the collinearity concern in the cautions register. It uses no parameters chosen by the author.

No preferred scenario. No scenario is designated as correct, primary, or best. Results are reported as an ensemble. Where a single surface is required for illustration, RES-04 (balanced evidence) is used and labelled as illustrative, not preferred.

Transformations. As specified per scenario in the register. Resistance values are relative ranks, not empirical coefficients, and are described as such throughout.

RES-06 and the RES-04 illustrative designation confirmed. Both were fixed before any resistance surface was produced.

10 Connectivity modelling

Effective distance and corridors. ArcGIS Pro Distance Accumulation and Optimal Region Connections, cores as sources, per scenario and per snapshot.

Current flow. Circuitscape.jl in **pairwise** mode, all core pairs, with cores as sources and grounds. Pairwise rather than omnidirectional because cores are defined a priori; an omnidirectional formulation answers a different question and is not used.

Normalisation. Cumulative current is normalised by the number of core pairs contributing, so that snapshots and core definitions yielding different core counts remain comparable. The normalisation formula is fixed here and applied identically everywhere.

Corridor definition. Continuous current surfaces summarised by quantile. Thresholded corridor polygons are used only for cartography and are never the basis of a statistical comparison.

Why this must be fixed now. The corridor definition sets the denominator for the protected-area test in §13. Choosing it after seeing the overlap result would let the analyst select the definition that produces the desired protection gap.

High-current definition. The **top decile** of normalised current within the analysis extent.

11 Network metrics

Cores are nodes; effective distances between core pairs are edge weights. Reported metrics:

- Probability of connectivity (PC) and per-node and per-link importance (dPC), computed in Conefor or an equivalent implementation.
- Betweenness centrality on the same graph, used for H1.

Dispersal distance parameter. PC requires a dispersal probability function. Median dispersal probability is set at 100 km, with sensitivity at **50, 200 and 400** km. No single value is reported alone.

100 km confirmed as an order-of-magnitude central value. No published cheetah natal-dispersal telemetry study exists for this system. The best available proxy is Melzheimer et al. 2020, who report communication hubs spaced 22.9 km (± 4.0 km) apart (independently corroborated at approximately 20 km in the Serengeti and Maasai Mara), with floaters visiting an average of 3.3 ± 1.9 hubs — implying individual movement typically spans on the order of several tens to 100 km within an extended range. This supports 100 km as a defensible order-of-magnitude central value, while acknowledging it is derived from hub-spacing data rather than a dedicated dispersal study. Sensitivity at 50, 200 and 400 km is reported alongside it; no single value is asserted as definitive.

12 Temporal comparison

Change is computed between consecutive snapshots and between the first and last snapshot, using identical cores, scenarios, resolution and normalisation. Both absolute and proportional change are reported. Change is attributed to covariate change by construction; no causal language is used.

13 Protected-area analysis

Source. A single named WDPA monthly release, recorded here on locking. **Release:** _____

Geometry handling. Valid polygons only. Point records are excluded and the count of exclusions is reported. Status and designation fields are filtered to **designated, inscribed and established**; proposed sites are excluded.

Predeclared null. The observed share of top-decile current area falling outside protected areas is compared against a null generated by **999 area-standardised spatial permutations of the protected-area configuration within the analysis extent**, preserving total protected area and patch size distribution.

Primary comparison denominator. The modelled network (the union of the top quartile of current), not the full analysis extent. The full extent is reported as a secondary comparison.

Rationale. Comparing against the full extent would largely recover the known fact that protected areas cover a minority of the landscape. Comparing against the modelled network asks the sharper question: within the area the model identifies as connectivity-relevant, is the highest-value subset disproportionately unprotected?

Significance threshold. $\alpha = 0.05$, reported with the observed effect size, not as a bare p -value.

Threshold sensitivity. The comparison is repeated at the top 5%, 10% and 20% of current. A result that holds only at one threshold is reported as threshold-dependent.

14 Sensitivity grid and the definition of robustness

Full grid at 1 km: 6 resistance scenarios $\times$ 3 core definitions $\times$ 4 snapshots = **72 runs**.

Resolution sensitivity: the same grid at 2 km and 5 km for the **first and last snapshots only** ($6 \times 3 \times 2 \times 2 = 72$ additional runs), giving **144 runs total**.

Rationale for restricting the resolution sensitivity. Repeating all four snapshots at all three resolutions triples compute for little added inference. The first and last snapshots bracket the change being measured, which is what the resolution check needs to test.

Definition of robust. A cell is robust if it falls in the top decile of normalised current in **at least 80% of runs** within a given snapshot (that is, $k \geq 15$ of $n = 18$ scenario-core combinations at 1 km).

Rationale for $k/n = 0.8$. The threshold is set high because the scenarios share input layers and will therefore agree more than genuinely independent models would. A permissive threshold would report shared-input agreement as biological robustness. This number is fixed before any agreement map is produced.

Reporting rule. Cells with agreement below the threshold are reported as *uncertain*, never as priorities, and appear in the monitoring index rather than in the corridor recommendations.

$k/n = 0.8$ confirmed. This is the single most consequential number in the protocol and the one most tempting to relax later. If it is relaxed after seeing the agreement map, it must appear in the amendment log with the pre- and post-values and the original result reported alongside.

15 Independent plausibility assessment

Route: plausibility check only. No resistance surface is fitted to the occurrence data. Consequently there is no fitted model to validate, and the word “validation” is prohibited in describing this analysis (see the claim dictionary, §17).

Evidence used. The archived Limpopo cheetah tracking dataset (nine individuals, 2003–2008) and published camera-trap occurrence evidence from central-eastern Namibia, both external to the core definition.

Test. Observed locations are compared against availability-matched random points drawn within the same landscape, asking whether observed locations fall in lower-resistance and higher-current cells more often than expected. Autocorrelation within tracks is accounted for by **subsampling to one location per individual per 24 hours**.

Declared limits. Nine individuals in one landscape twenty years ago cannot speak to a four-country network in 2024. A positive result is localised support only. A negative or null result is reported with equal prominence and does not invalidate the modelling, but does bound the discussion.

16 Monitoring priority index

Reporting unit. 10 km cells, matching the resolution of the occurrence evidence rather than the modelling resolution.

Components, each normalised to $[0, 1]$ by min-max scaling within the extent: (i) connectivity importance (normalised current and dPC); (ii) magnitude of connectivity change 2012–2024; (iii) model uncertainty (inverse of ensemble agreement); (iv) evidence age; (v) evidence sparsity; (vi) threat exposure (change in human pressure).

Predeclared weight sets.

Set	Connectivity	Change	Uncertainty	Evid. age	Evid. sparsity	Threat
W1 (balanced, headline)	0.25	0.20	0.15	0.15	0.15	0.10
W2 (connectivity-led)	0.45	0.15	0.10	0.10	0.10	0.10
W3 (knowledge-gap-led)	0.15	0.15	0.20	0.25	0.20	0.05

Stability testing. Spearman rank correlation between weight sets across all cells, and Jaccard overlap of the top 20 cells between weight sets. A cell enters the recommended tier only if it appears in the top 20 under **all three** weight sets.

Transparency requirement. All six component scores are published per cell alongside the composite, so any reader can reweight and obtain their own ranking. The index is decision support under stated assumptions, not an objective ordering.

The three weight sets are confirmed and fixed. If a partner organisation later supplies operational weights, those are added as a fourth set in an amendment, not substituted for these.

17 Claim dictionary

Permitted and prohibited language, binding on the manuscript.

Prohibited	Required instead
functional corridor; cheetah corridor	area of high modelled potential structural connectivity
cheetahs move through / use this area	this area is a modelled structural priority warranting field validation
validation; validated	independent plausibility assessment; consistent with
conflict hotspot	area of high modelled livestock exposure
population decline; caused	spatially coincided with; associated with
current cheetah range	historical confirmed-range evidence, 2010–2016
absent; no cheetahs	no records; unknown detection and effort
the corridor network	the modelled network under scenario X
this resistance surface is correct	results were evaluated across a predeclared ensemble

Every prohibited term is checked by text search before submission. This check is a required item in the pre-submission audit.

18 Decision and stopping rules

If corridor rankings are unstable (no robust set emerges under §14), the paper’s primary output becomes the uncertainty-guided monitoring design, and the corridor map is demoted to a supporting figure. This is a legitimate result, not a failure, and is declared here so it cannot be presented as the plan all along or quietly avoided.

If the protected-area result is not significant, H2 is reported as unsupported and the discussion addresses why, without threshold shopping.

If the plausibility assessment is null, it is reported in full and the discussion is bounded accordingly.

All scenarios are retained and reported. No scenario is dropped for producing inconvenient or uninteresting results.

19 Reproducibility and data handling

Scripting. All geoprocessing is scripted in arcpy. ModelBuilder is used for exploration only; no result in the manuscript derives from an unscripted step.

Version recording. ArcGIS Pro, Julia, Circuitscape.jl, R or Python, and every dataset version and access date are recorded in the data manifest.

Redistribution limits. Raw WDPA data are not redistributed; users obtain them independently under Protected Planet terms. Occurrence data are shared only at the resolution the source archive permits. Derived summaries and all code are published.

Archive. Code, protocol, manifest and derived non-restricted outputs are deposited with a DOI on completion.

20 Amendment log

Every change to a locked value after the sign-off date is recorded here before the change is made. The log is published with the manuscript.

No.	Date	Section and value	Changed from / to	Reason, and whether any result had been seen
1				
2				
3				
4				

The final column is the one that matters. An amendment made before any relevant result was viewed is routine. An amendment made afterwards is not disqualifying, but it must be visible, and the pre-amendment result is reported alongside the post-amendment result.